STATISTICAL MECHANICS

3

- The system will evolve accoring to Hamiltonion dynamics, $x = (q_{1:N}, p_{1:N}) \in \mathbb{R}^{6N} = \mathbb{X}$

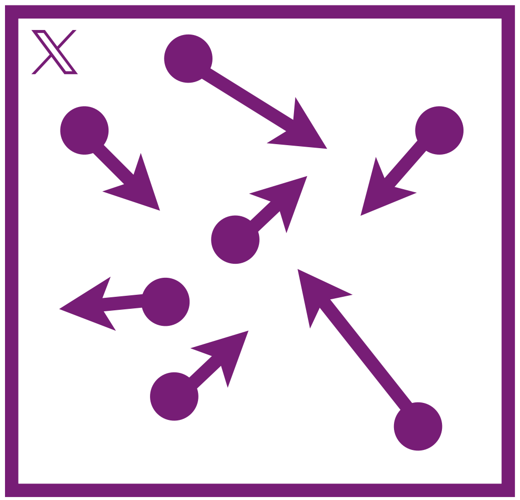
$$H(x) = \sum_i V(q_i) + \frac{1}{2m} \sum_i |p_i|^2$$

- Its infeasible to track every particle over time when N is large (e.g. $N = 10^{23}$)

- The stationary distribution of the particles satisfies :

$$\eta(x) \propto \exp(-H(x))$$

- Suppose we have a gas of N particles in a box in $\mathbb{R}^3$ where particle i has mass m and velocity v_i
  - A particle i has mass m with position q_i and momentum p_i



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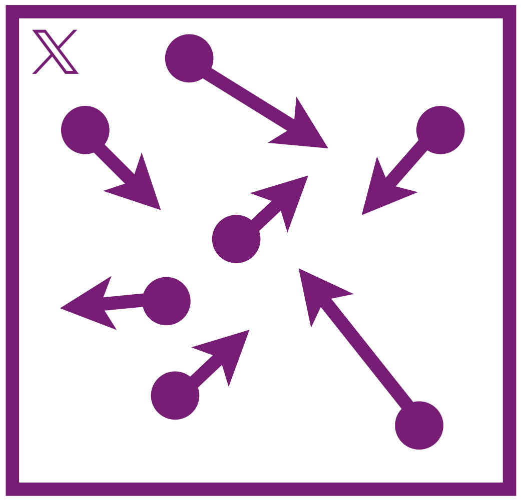

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